Individual assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
```

```
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
      here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
      sheet = "Water Quality",
      na = c("", "n/a", "N/A", "over", "173+")
)
```

Cleaning

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
```

```r
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

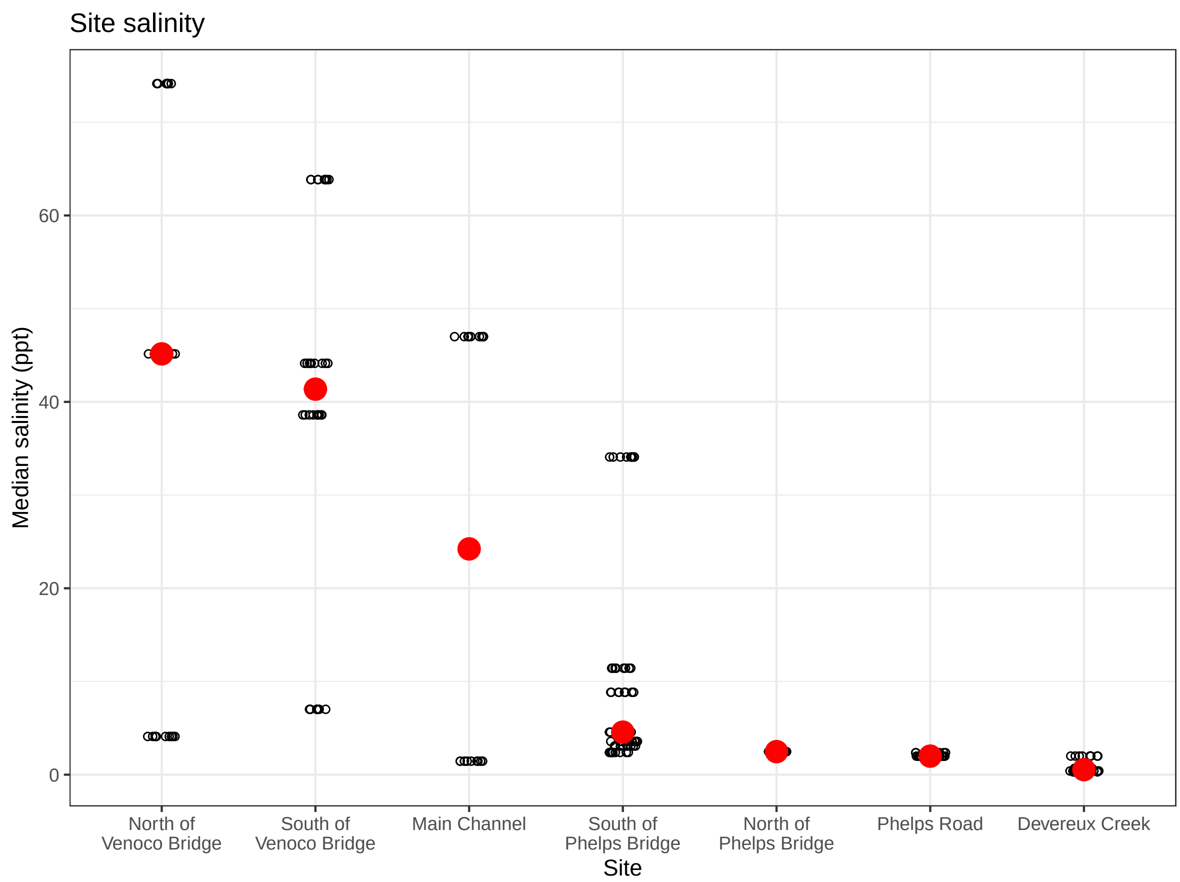
```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
```

```r
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(
    site_full, med_sal, .fun = "median", .na_rm = TRUE
    ),
         site_full = fct_rev(site_full))
```

Class visualizations

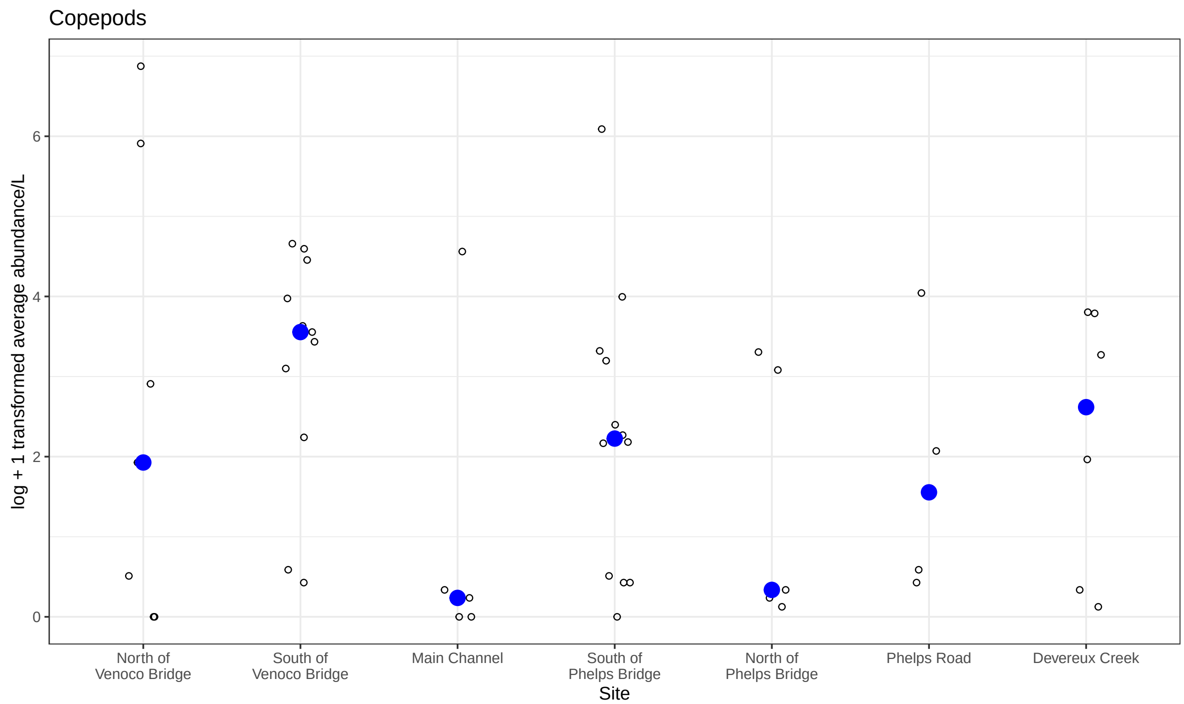
```r
# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
```

```r
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()

# displaying object
salinity
```




```r
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

```
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()

# displaying object
copepods
```



1. Visualizing differences across sites in major taxon abundance

a.

- x-axis: site
- y-axis: log + 1 transformed average ostracod abundance/L
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()` with a visually distinct point

b.

```
# creating new object from clean aquatic inverts
ostracods <- aq_ins_clean |>
  # filter to only include ostracods
  filter(taxon == "ostracod") |>
  # create log + 1 transformed abundance column
  mutate(log_ave_lit = log(ave_lit + 1))
```

```r
# display 10 random rows from data frame
slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6      NA     NA      NA
 2 2022-05-11_NDC  2022-05-11 00:00:00 NDC   ostr~   0       9.11   0.398   9.4
 3 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1  ostr~   0      NA     NA      NA
 4 2024-01-28_NPB  2024-01-28 00:00:00 NPB   ostr~   3.33    7.43   3.08    6.6
 5 2022-05-05_NEC  2022-05-05 00:00:00 NEC   ostr~   0.267   9.45  63.8     7.3
 6 2022-01-28_NDC  2022-01-28 00:00:00 NDC   ostr~   9.47    7.65   2.00    9.55
 7 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533  NA     47      14.0
 8 2023-02-12_NEC  2023-02-12 00:00:00 NEC   ostr~   2.13   NA     NA      NA
 9 2021-11-23_NPB2 2021-11-23 00:00:00 NPB2  ostr~   0.267   7.59   1.99    0.65
10 2022-11-12_NEC  2022-11-12 00:00:00 NEC   ostr~   0       7.33   7.01    2.4
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c.

```r
# base layer
ostracods_plot <- ggplot(data = ostracods,
                         # x-axis: site
                         # y-axis: log transformed average abundance/L
                         mapping = aes(x = site_full,
                                       y = log_ave_lit)) +
  # first layer: points to represent individual observations
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # second layer: larger colored diamonds to represent site medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               shape = 23,
               color = "darkorange") +
  # wrap long site names so x-axis labels are readable
```

```
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # relabel axes and add a descriptive title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # use a non-default theme
  theme_bw()

# display plot
ostracods_plot
```